🏛️ Strategic Recovery & Capital Optimizer (PoC)

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Target Stakeholders: Chief Risk Officers (CRO), CFOs, & Basel IV Steering Committees

📊 Executive Summary

The **Strategic Recovery & Capital Optimizer** is an executive-grade decision-support portal designed to navigate the complexities of the modern regulatory landscape. By unifying recovery operations with capital adequacy metrics (CET1), this platform demonstrates how high-performance recovery strategies directly mitigate **Capital Drag** and drive shareholder value.

Strategic Business Value:

- **CET1 Ratio Optimization:** Simulates the reduction of Risk-Weighted Assets (RWA) through increased recovery yields, effectively "freeing up" Tier 1 capital for reinvestment.
- **ROTCE Pathfinding:** Provides a direct line of sight toward **Return on Tangible Common Equity (ROTCE)** targets (e.g., 11% by 2026), proving that recovery efficiency is a primary driver of bottom-line profitability.
- **Macro-Resilience:** Stress-tests Net Charge-Off (NCO) projections across a 9-quarter horizon against "Severely Adverse" economic scenarios.

🚀 Strategic Modules & Executive UX

The portal features a "Human-in-the-Loop" architecture, protected by an **Executive Gateway** for secure leadership review.

1. Capital Report & WARR Analysis

- **Goal:** Track the yield health of Branded Cards and Retail segments.

- **Analytics:** Dynamic modeling of **Weighted Average Recovery Rates (WARR)** with automated Year-over-Year (YoY) variance tracking to identify performance leakage and its impact on CET1 ratios.

2. Dynamic Stress Testing (9-Quarter Engine)

- **Goal:** Benchmark institutional resilience against peer institutions (e.g., JPM, Amex).
- **Simulation:** An NCO projection engine that overlays macro-economic severity sliders (Unemployment, GDP), allowing leaders to visualize capital requirements in recessionary environments.

3. Vendor Governance & Placement Simulator

- **Sunburst Analytics:** Visualizes the relationship between Tiered vendors and **Capital Drag (bps)**.
- **Inventory Swap Simulator:** A "what-if" tool to calculate the net recovery gain of shifting inventory between agencies or moving to legal-track recovery.
- **Jurisdictional Heatmap:** A Choropleth-based tracker monitoring state-level regulatory exceptions and audit updates (e.g., CPPA audits in CA).

🛠️ Technical Architecture

This project demonstrates proficiency in financial engineering, high-fidelity data visualization, and secure web deployment.

Layer	Technical Implementation
UI/UX Framework	`Streamlit` with custom CSS-in-Python styling for an "Executive Gateway" interface.
Financial Engine	High-performance `@st.cache_data` logic for simulating CET1 basis point (bps) impacts across multi-million row dataframes.
Visualizations	`Plotly Graph Objects` for financial curves and `Plotly Express` for sunburst/geospatial analytics.

📊 Strategic KPI Glossary

Key metrics calculated within the PoC are aligned with standard banking and recovery operations:

- **CET1 Ratio:** The primary measure of a bank's financial strength; recovery lifts reduce RWA to boost this ratio.
- **ROTCE:** Return on Tangible Common Equity; the ultimate yardstick for shareholder value and profitability.
- **Capital Drag (bps):** The "cost of carry" for inventory measured in basis points, reflecting capital tied up during long liquidation timelines.
- **Yield Premium:** The additional recovery percentage gained by utilizing higher-cost legal channels over standard collections.

💻 Installation & Usage

To launch the decision engine locally:

1. **Clone the repository:**
   ```
   git clone https://github.com/your-username/strategic-recovery-optimizer.git
   ```
2. **Install dependencies:**
   ```
   pip install -r requirements.txt
   ```
3. **Run the application:**
   ```
   streamlit run app.py
   ```

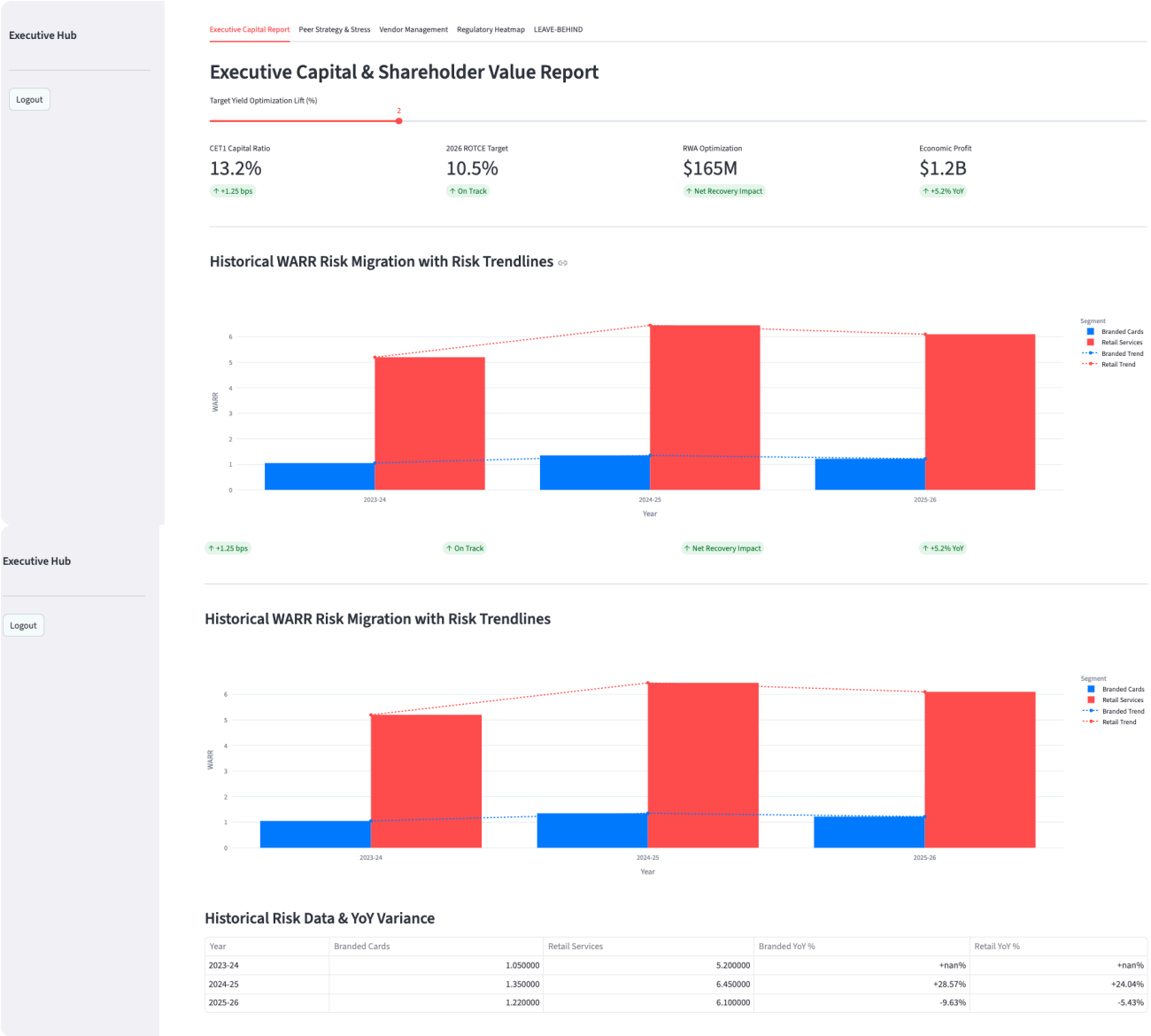
Executive Gateway Access Code: `Recovery2026`

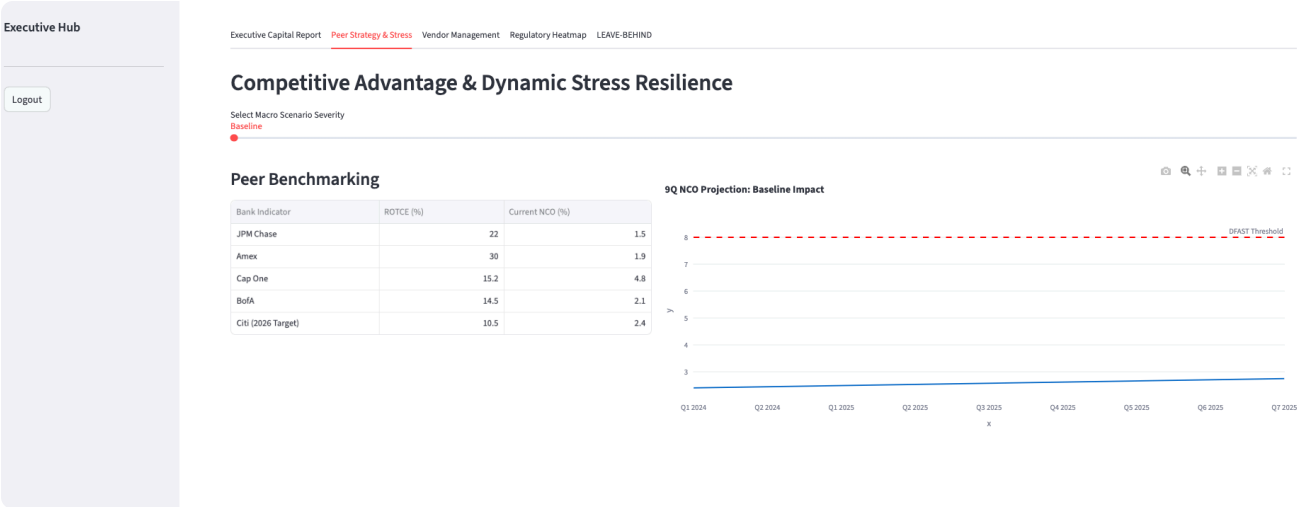
📁 Repository Strategy

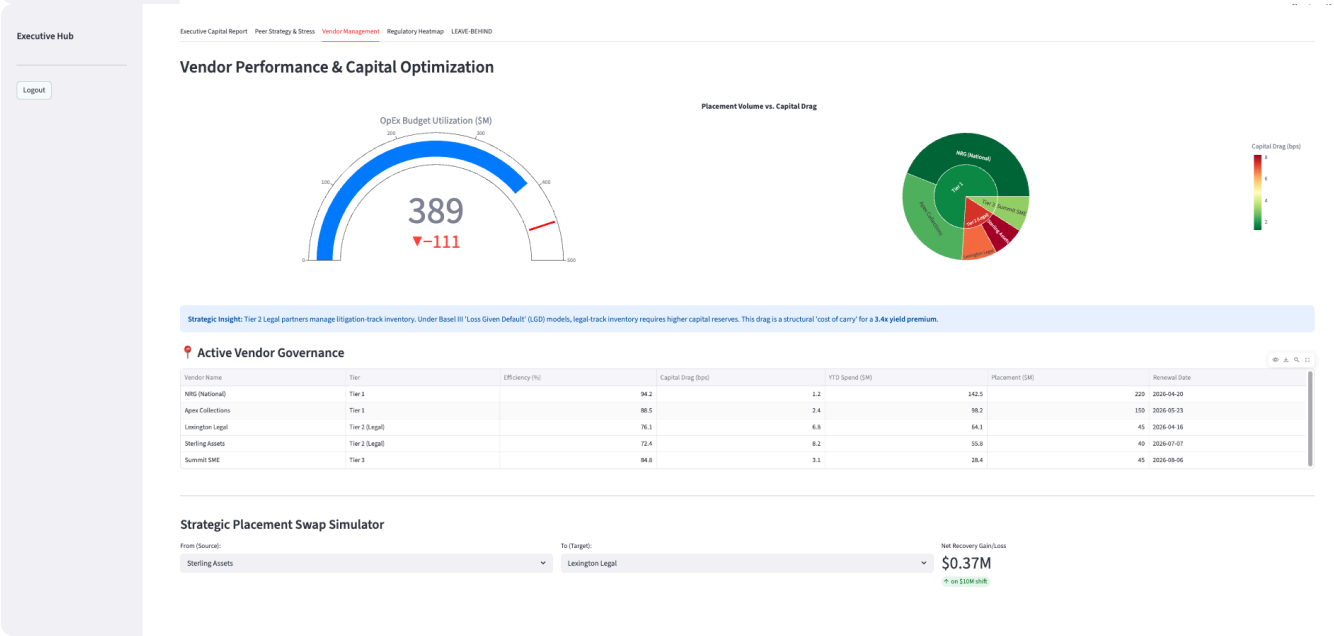
This repository serves as a **Narrative of Authority**, shifting the focus from simple operations to institutional capital management.

Operational Focus (Old)	Capital Strategy Focus (New)
Lists "Vendor Placement" as a feature.	Frames it as "Managing Capital Drag (bps)" for the CFO.

Screen Shots




Executive Hub
Logout
Executive Capital Report
Peer Strategy & Stress
Vendor Management
Regulatory Heatmap
LEAVE-BEHIND
Competitive Advantage & Dynamic Stress Resilience
Select Macro Scenario Severity
Baseline
Peer Benchmarking
Bank Indicator | ROTCE (%) | Current NCO (%)
JPM Chase | 22 | 1.5
Amex | 30 | 1.9
Cap One | 15.2 | 4.8
BofA | 14.5 | 2.1
Citi (2026 Target) | 10.5 | 2.4
9Q NCO Projection: Baseline Impact
DFAST Threshold
Q1 2024
Q2 2024
Q1 2025
Q2 2025
Q3 2025
Q4 2025
Q5 2025
Q6 2025
Q7 2025


Executive Hub
Logout
Executive Capital Report
Peer Strategy & Stress
Vendor Management
Regulatory Heatmap
LEAVE-BEHIND
Vendor Performance & Capital Optimization
OpEx Budget Utilization ($M)
389
▼−111
Placement Volume vs. Capital Drag
NRG (National)
Capital Drag (bps)
Strategic Insight: Tier 2 Legal partners manage litigation-track inventory. Under Basel III 'Loss Given Default' (LGD) models, legal-track inventory requires higher capital reserves. This drag is a structural 'cost of carry' for a 3.4x yield premium.
Active Vendor Governance
Vendor Name | Tier | Efficiency (%) | Capital Drag (bps) | YTD Spend ($M) | Placement ($M) | Renewal Date
NRG (National) | Tier 1 | 94.2 | 1.2 | 142.5 | 220 | 2026-04-20
Apex Collections | Tier 1 | 88.5 | 2.4 | 98.2 | 150 | 2026-05-23
Lexington Legal | Tier 2 (Legal) | 76.1 | 6.8 | 64.1 | 45 | 2026-04-16
Sterling Assets | Tier 2 (Legal) | 72.4 | 8.2 | 55.8 | 40 | 2026-07-07
Summit SME | Tier 3 | 84.8 | 3.1 | 28.4 | 45 | 2026-08-06
Strategic Placement Swap Simulator
From (Source):
Sterling Assets
To (Target):
Lexington Legal
Net Recovery Gain/Loss
$0.37M
on $10M shift

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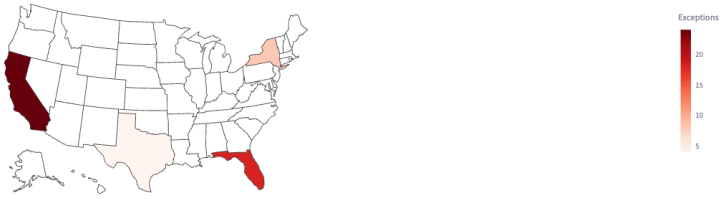
Jurisdictional Governance



📍 Risk Detail: Federal

Strategic Category	Detail
Jurisdiction	Federal
Risk Level	🟡 Moderate
Focus Area	OCC Oversight
Total Exception Volume	52 Active Exceptions
Status	Resource Review terminated Dec 2025.

Executive Hub

Logout

Executive Capital Report | Peer Strategy & Stress | Vendor Management | Regulatory Heatmap | LEAVE-BEHIND

Executive Leave-Behind

Strategic Positioning Summary (Q1 2026)

- **Capital Impact:** Yield optimization adds **1.25 bps** to CET1 headroom.
- **Historical Context:** Proactive 2024 optimization stabilized risk velocity in Branded Cards.

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